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Editor-in-Chief
IEEE Transactions on Aerospace and Electronic Systems

Dear Editor-in-Chief,

We submit our manuscript, “A Closed-Loop Subspace-RFS Framework for Multi-Target DOA Tracking,” for consideration as a regular paper in the IEEE Transactions on Aerospace and Electronic Systems.

The paper addresses a core TAES problem—multi-target direction-of-arrival (DOA) tracking with compact sensor arrays—and its contributions sit squarely in the Transactions’ scope of target tracking, random-finite-set (RFS) filtering, and radar systems:

- **A gated closed-loop architecture** that replaces the standard open-loop estimator-to-tracker cascade: the RFS tracker’s motion-compensated prediction is fed back through a two-stage validation gate to refine the DOA estimator’s signal subspace. The gate guarantees exact fallback to the open-loop estimate, so the deployed system never underperforms the cascade it replaces; at low SNR it lowers localization error by up to 56%, and under target motion it cuts track identity switches by 15%.
- **Underdetermined tracking capability:** with a higher-order-cumulant front-end (COP), a compact $M=8$ array tracks $K=10 > M-1$ targets, detecting significantly more targets than a MUSIC-PHD cascade. The back-end is interchangeable across track-oriented PHD, labeled multi-Bernoulli (LMB), and δ -GLMB filters, and a physics-based constant-acceleration motion model makes the δ -GLMB back-end best under maneuver.
- **Supporting guarantees:** a posterior Cramér–Rao bound for the complete closed loop (identifiability is restored exactly where the data-only bound diverges) and a minimum-variance subspace-fusion proof with a gate-enforced benefit condition.

The evaluation is Monte-Carlo throughout (up to 500 trials with 95% confidence intervals), against MUSIC, closed-loop MUSIC, LMB, and δ -GLMB baselines, and we explicitly delineate the operating regime of each ingredient rather than claiming uniform superiority.

This manuscript is original, has not been published, and is not under consideration by any other journal. Thank you for considering our work.

Sincerely,

Jin Ho Choi